

Milestone 2 – Hash Table, Due September 23, 2026 4:00 PM Pacific Time

The following files will be provided to you, for completion of your milestone:

- hash_node.h // header file containing hash node structure
- hash_node.cpp // cpp file containing hash node implementation
- hash_table.h // header file containing hash table class
- json.hpp // header file for processing json files
- milestone2.h // header file for milestone2 method declarations
- milestone2.json // json file containing test cases and its transactions
- milestone2_config.json // json configuration (properties) file
- outputFile - testcase 1.txt // expected output for testcase 1
- milestone2.cpp /* cpp file containing main, which does the following:

- Reads configuration file (json format) to:
 - retrieve inputFile (test case file (json format))
 - retrieve outputFile (text file containing generated output)
 - retrieve errorLogFile (text file containing error messages)
- process inputFile test cases
- write output to outputFile */

1. Write a Unified Modeling Language (UML) Class Diagram and Sequence Diagram for this milestone.
2. Write a basic hash table implementation, which uses the files listed above, and includes the following in a separate cpp file:

- hash_table.cpp – implementation file that contains the following methods:
 1. getTable - return the hash table
 2. getSize - return the size of the hash table
 3. calculateHashCode – perform hashing function
 4. isEmpty - Check if the HashTable is empty
 5. getNumberOfItems - return number of items in the hash table
 6. add - adds a new node to the hash table
 7. remove – remove node with key value
 8. clear - remove all entries from the table
 9. getItem - returns pointer to the HashNode
 - 10.contains - check if a node with key exists in the table
 - 11.printTable - print out contents of hash table to console and output file

The total number of points for this milestone is **110**, which will be based upon the following:

- All file(s) are submitted via Canvas (5)
- Each submitted/modified file must have student's name (-10% of total milestone points if missing)
- The UML Class Diagram and Sequence Diagram are documented (20)
- Each submitted/modified file must include description of changes made to a program, and its change date (1)
- Program compiles with all of the provided files (1)
- The following methods are documented:
 1. getTable - return the hash table (1)
 2. getSize - return the size of the hash table (1)
 3. calculateHashCode - perform hashing function (1)
 4. isEmpty - Check if the HashTable is empty (1)
 5. getNumberOfItems - return number of items in the hash table (1)
 6. add - adds a new node to the hash table (1)
 7. remove - remove node with key value (1)
 8. clear - remove all entries from the table (1)
 9. getItem - returns pointer to the HashNode (1)
 - 10.contains - check if a node with key exists in the table (1)
 - 11.printTable - print out contents of hash table to console and output file (1)
- The following methods run without errors:
 1. getTable - return the hash table (2)
 2. getSize - return the size of the hash table (2)
 3. calculateHashCode - perform hashing function (2)
 4. isEmpty - Check if the HashTable is empty (2)
 5. getNumberOfItems - return number of items in the hash table (2)
 6. add - adds a new node to the hash table (2)
 7. remove - remove node with key value (2)
 8. clear - remove all entries from the table (2)
 9. getItem - returns pointer to the HashNode (2)
 - 10.contains - check if a node with key exists in the table (2)
 - 11.printTable - print out contents of hash table to console and output file (2)
- The following test cases are processed, and produce expected output (10 per test case; 50 total)
- Extra Credit - use industry standard test program and/or extract test cases, in separate json test file